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***B.Tech. Degree VIII Semester Supplementary Examination
September 2016***

**ME 1801 COMPRESSIBLE FLUID FLOW
(2012 Scheme)**

(Approved Gas Tables are permitted)

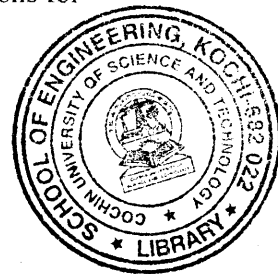
Time : 3 Hours

Maximum Marks : 100

PART A
(Answer *ALL* questions)

(8 × 5 = 40)

- I. (a) Briefly explain the propagation of waves from a source of disturbance moving at supersonic speed.
- (b) Classify nozzle and diffuser in terms of the expression
- $$\frac{dA}{A} = \frac{-dp}{\rho c^2} [1 - M^2].$$
- (c) What is meant by a shock wave? Shock waves cannot develop in a subsonic flow. Why?
- (d) Explain two limiting cases of oblique shock waves.
- (e) What is Fanno flow? What are the assumptions made in deriving equations for Fanno flow? Give practical examples.
- (f) Describe Rayleigh line with a neat sketch.
- (g) Briefly explain temperature recovery factor.
- (h) Explain the working of Schlieren apparatus.



PART B

(4 × 15 = 60)

- II. A converging diverging nozzle has an exit area to throat area ratio of 2. Air enters the nozzle with a stagnation pressure of 7 bar and a stagnation temperature of 100°C. The throat area is 6.5 cm². Determine the exit pressure, exit temperature, exit Mach number and exit velocity for the following conditions. (15)
- (i) Sonic velocity at the throat; diverging section acting as a Nozzle.
- (ii) Sonic velocity at the throat; diverging section acting as a Diffuser.

OR

(P.T.O.)

- III. (a) Derive an expression to indicate the effect of Mach number on compressibility. (7)
 (b) An aircraft is flying at an altitude of 11000 m, at 800 km/hr. The air is reversibly compressed in an inlet diffuser. The inlet temperature is 216.65 K and the pressure is 0.226 bar. If the Mach number at the exit of the diffuser is 0.35, calculate the following: (8)
 (i) Entry Mach number.
 (ii) Velocity, pressure and temperature of air at the diffuser exit.

- IV. Derive (i) Prandtl-Meyer relation and (ii) Rankine Hugoniot relation for normal shock waves. (15)

OR

- V. An oblique shock wave occurs at leading edge of a symmetrical wedge. Air flow has a Mach number of 2 and deflection angle of 15° . Determine the following for strong and weak waves. (15)
 (i) Wave angle.
 (ii) Pressure ratio.
 (iii) Temperature ratio
 (iv) Downstream Mach number.

- VI. A circular duct passes 10 kg/s of air at an exit Mach number of 0.5. The entry pressure and temperature are 3.6 bar and 370°C respectively and co-efficient of friction is 0.004. If the Mach number at entry is 0.15, determine (15)
 (i) Diameter of the duct.
 (ii) Length of the duct.
 (iii) Pressure and temperature at exit.
 (iv) Stagnation pressure loss.

OR

- VII. A combustion chamber in a gas turbine plant receives air at 340 K, 0.5 bar and 70 m/s. The air fuel ratio is 29 and the calorific value of the fuel is 41.87 MJ/kg. Taking $R = 0.287 \text{ kJ/kg-K}$ for the gas, determine (15)
 (i) Initial and final Mach number.
 (ii) Final pressure, temperature and velocity of gas.
 (iii) Stagnation pressure loss in combustion chamber.
 (iv) Maximum stagnation temperature attainable.

- VIII. (a) Derive the Rayleigh Supersonic Pitot formula. (5)
 (b) What is hot wire anemometer? Explain two arrangements of hot wire anemometer with neat sketches. (10)

OR

- IX. With neat figures, explain the working of : (15)
 (i) Shock tube.
 (ii) Velocimeter.
 (iii) Supersonic Wind Tunnel.